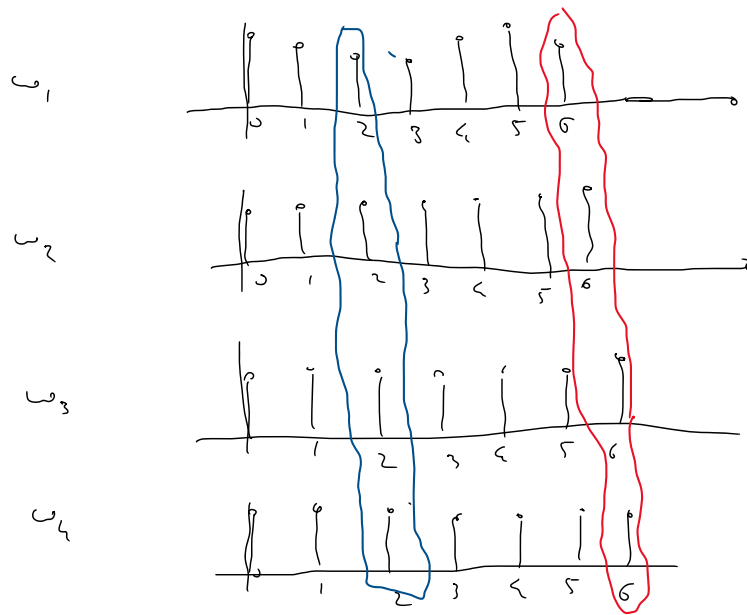


$$x_i = x(t_i)$$

$$[x(t_1) \ x(t_2) \ \dots \ x(t_n)]$$



$$[X(10, \omega_i) \ X(13, \omega_i)]$$

$$[X(11, \omega_i) \ X(14, \omega_i)]$$

$$[X(10, \omega_i) \ X(13, \omega_i) \ X(20, \omega_i)]$$

$$[X(11, \omega_i) \ X(14, \omega_i) \ X(21, \omega_i)]$$

$$[X(12, \omega_i) \ X(15, \omega_i) \ X(22, \omega_i)]$$

$$[X(m_1, \omega_i) \ X(m_2, \omega_i) \ \dots \ X(m_n, \omega_i)]$$

$$[X(m_1+1, \omega_i) \ X(m_2+1, \omega_i) \ \dots \ X(m_n+1, \omega_i)]$$

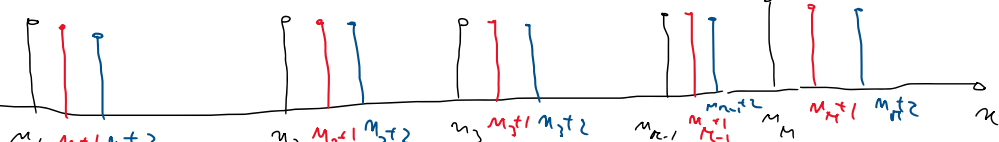
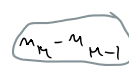
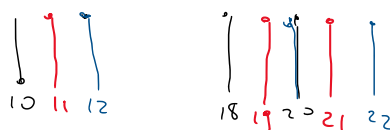
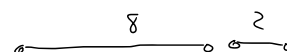
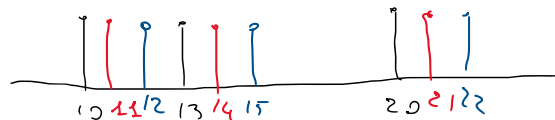
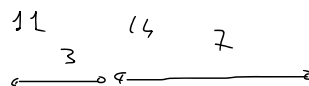
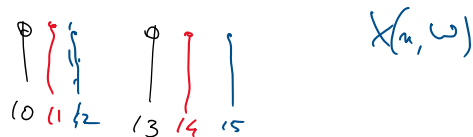
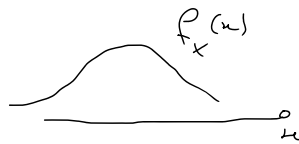


Diagram illustrating the construction of a sequence of functions $X(n_i + 1, \omega_i)$ for $i=1$ to M . The diagram shows a horizontal axis with points $n_1, n_2, n_3, \dots, n_{M-1}, n_M$. Above the axis, there are vertical lines representing the functions. The first line is labeled $X(n_1 + 1, \omega_1)$ and the last line is labeled $X(n_M + 1, \omega_M)$. The axis is labeled with $n_1, n_2, n_3, \dots, n_{M-1}, n_M$, and n . The functions are defined on intervals $[n_i, n_{i+1}]$ for $i=1$ to $M-1$, and $[n_M, n]$ for $i=M$. The functions are labeled $X(n_i + 1, \omega_i)$ for $i=1$ to M .

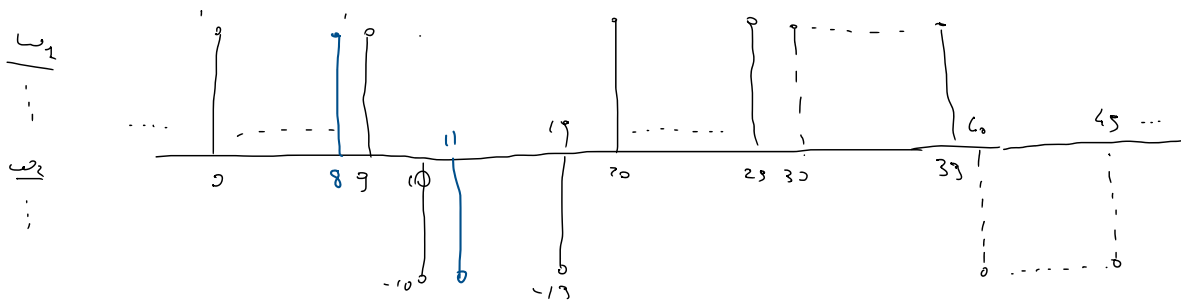
$$\{x_1, x_2, \dots, x_n\}$$

$$\{P_1, P_2, \dots, P_n\}$$



$$X(m, \omega)$$

$$\begin{cases} 1 & \text{con prob } p \\ 0 & = = 1-p \end{cases}$$



$$(X(0, \omega_i), X(3, \omega_i)) \in \{(1, 1), (-1, -1)\}$$

$$P[(X(0), X(3)) \in (1, 1)] = \frac{1}{2}$$

$$P[X(2), X(3)] = (-1, -1) = \frac{1}{2}$$

$$f(x(0), x(3)) = (1, -1) = 0$$

$$\vec{r} = (-1, 2) = \vec{0}$$

$$P[X(0) = 1] = \frac{1}{2}$$

$$P[X(3)=1] = \frac{1}{2}$$

$$12(x(0) = -1) = \frac{1}{2}$$

$$P\{X(3) = -1\} = \frac{1}{2}$$

$$P[(X(8), X(11)) = (1, 1)] = P[X(8) = 1] P[X(11) = 1] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P[x(8), x(11)] = (-1, -1) = \frac{1}{4}$$

$$P[X(n) = 1] = \frac{1}{2}$$

$$P[(X(8), X(11)) = (-1, 1)] = \frac{1}{4}$$

$$P[X(u) = -1] = \frac{1}{2}$$

$$P[(X(8), X(11)) = (1, -1)] = \frac{1}{4}$$

$$[X(n_1) \quad X(n_2) \dots X(n_M)]$$

$$\begin{bmatrix} x(n_1+1) & x(n_2+2) & \dots & x(n_{n_2}+1) \end{bmatrix}$$

$$[X(n_1+z) \quad X(n_2+z) \quad \dots \quad X(n_n+z)]$$

$$m_2 - m_1, m_3 - m_2, \dots, m_n - m_{n-1}$$

$$\underbrace{n_2 - n_1}, \underbrace{n_3 - n_1}, \dots, \underbrace{n_n - n_1}$$

$$n_1+h \quad n_2+h \quad \dots \quad n_M+h \quad (n_2+h)-(n_1+h), (n_3+h)-(n_2+h), \dots, (n_M+h)-(n_{M-1}+h)$$

$$X(n, \omega)$$

$$[X(n_1, \omega) \quad X(n_2, \omega) \quad \dots \quad X(n_M, \omega)]$$

$$f_X(d_1, d_2, \dots, d_M; n_1, n_2, \dots, n_M) = f_X(d_1, d_2, \dots, d_M; n_1+h, n_2+h, \dots, n_M+h) \quad \forall h$$

$$[X(n_1+h, \omega) \quad X(n_2+h, \omega) \quad \dots \quad X(n_M+h, \omega)]$$

$$[X(n_1, \omega) \quad X(n_2, \omega) \quad \dots \quad X(n_{M-1}, \omega)] \rightarrow f_X(d_1, \dots, d_{M-1}; n_1, n_2, \dots, n_{M-1}) = \int_{-\infty}^{+\infty} f_X(d_1, d_2, \dots, d_M; n_1, n_2, \dots, n_M) dd_M$$

$$[\cancel{X(n_1+h, \omega)} \quad \cancel{X(n_2+h, \omega)} \quad \dots \quad \cancel{X(n_{M-1}+h, \omega)}] \rightarrow f_X(d_1, \dots, d_{M-1}; n_1+h, n_2+h, \dots, n_{M-1}+h) = \int_{-\infty}^{+\infty} f_X(d_1, d_2, \dots, d_M; n_1+h, n_2+h, \dots, n_{M-1}+h, n_M+h) dd_M$$

$$[X(n_1, \omega) \quad X(n_2, \omega) \quad \dots \quad X(n_M, \omega)] \quad P[X(n_1)=x_1, X(n_2)=x_2, \dots, X(n_M)=x_M] \quad \forall (x_1, x_2, \dots, x_M)$$

$$[X(n_1+h, \omega) \quad X(n_2+h, \omega) \quad \dots \quad X(n_M+h, \omega)] \quad P[X(n_1+h)=x_1, X(n_2+h)=x_2, \dots, X(n_M+h)=x_M] \quad \forall (x_1, x_2, \dots, x_M)$$

$$[X(n_1, \omega) \quad X(n_2, \omega), \dots, X(n_M, \omega)] =$$

$$P[X(\underbrace{n_1}_{n_1+h})=x_1, X(\underbrace{n_2}_{n_2+h})=x_2, \dots, X(\underbrace{n_M}_{n_M+h})=x_M] = \sum_{x_M \in A} P[\dots | X(n_M)=x_M] P[X(n_M)=x_M]$$

$$= \sum_{x_M} P[X(\underbrace{n_1}_{n_1+h})=x_1, X(\underbrace{n_2}_{n_2+h})=x_2, \dots, X(\underbrace{n_M}_{n_M+h})=x_M]$$

$$\begin{bmatrix} \sigma_1^2 & \text{cov}(x_1, x_2) \\ \text{cov}(x_1, x_2) & \sigma_2^2 \end{bmatrix}$$

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} \quad \underline{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} \quad \mu_i = E[X_i]$$

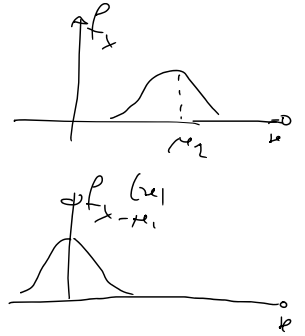
$$\underline{X} - \underline{\mu} = \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \\ \vdots \\ X_n - \mu_n \end{bmatrix}$$

$$f(x) = f\left(x + \underbrace{\mu_1}_{\text{shift}}\right)$$

↑
f_x

$$\{x_n - \mu_n\}$$

$$(X - \mu)(X - \mu)^T = \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \\ \vdots \\ x_n - \mu_n \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 & x_2 - \mu_2 & \dots & x_n - \mu_n \end{bmatrix}$$



$$= \left[(x_i - \mu_i)(x_j - \mu_j) \right]_{\{i,j\}}$$

$$\underline{C}_X = \left[\underbrace{E[(x_i - \mu_i)(x_j - \mu_j)]}_{\substack{\text{cov}(x_i, x_j) \text{ se } i \neq j \\ \text{VAR}(x_i) \text{ se } i = j}} \right]_{\{i,j\}} = E[(X - \mu)(X - \mu)^T]$$

$$[X(n_1 + h), X(n_2 + h)]$$

$$E[X(n_1 + h)X(n_2 + h)]$$

$$E[XY] = \sum_x \sum_y \underbrace{xy}_{\text{dip. solo da } n_2 - n_1} P[X=x, Y=y]$$

$$= \sum_{x_1} \sum_{x_2} x_1 x_2 \underbrace{P[X(n_1 + h) = x_1, X(n_2 + h) = x_2]}_{\text{dip. solo da } n_2 - n_1}$$

se il processo è stazionario
allora

$$E[X(k)X(m)]$$

dip. solo da $k-m$

statistica
correlazione in
tempo-tempo



$$E[X(k)X(k-m)]$$

dip. solo da m

statistica
correlazione in tempo-velocità



$$E[X(n_1)X(n_2)] = R_X(n_2 - n_1)$$





stazionari nelle medie

$$[X(n_1, \omega) \quad X(n_2, \omega) \quad \dots \quad X(n_m, \omega)]$$

$$\underline{X} = \begin{bmatrix} X(n_1, \omega) \\ X(n_2, \omega) \\ \vdots \\ X(n_m, \omega) \end{bmatrix}$$

$$\underline{\mu}_X = \begin{bmatrix} E[X(n_1)] \\ E[X(n_2)] \\ \vdots \\ E[X(n_m)] \end{bmatrix}$$

$$= \begin{bmatrix} E[X(n_1+h)] \\ E[X(n_2+h)] \\ \vdots \\ E[X(n_m+h)] \end{bmatrix} \quad \forall h$$

$$\underline{C}_X = E \left[\begin{pmatrix} X(n_1+h) \\ X(n_2+h) \\ \vdots \\ X(n_m+h) \end{pmatrix} \begin{pmatrix} X(n_1) \\ X(n_2) \\ \vdots \\ X(n_m) \end{pmatrix} \right] = [Cov(X(n_i), X(n_j))]$$

$$= \left[E \left((X(n_i) - \mu_i) (X(n_j) - \mu_j) \right) \right] = \underline{C}_{i,j} \sigma_X^2$$

$$VAR[X(n_i)] = VAR[X(n_j)] \quad \text{STAZ. in senso lato}$$

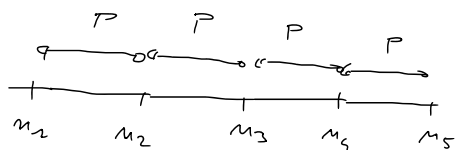
$$\sigma_X^2 = VAR[X(n_i)] = E[X^2(n_i)] - \{E[X(n_i)]\}^2 =$$

$$= E[X(n_i) X(n_i)] - \{E[X(n_i)]\}^2$$

$$= R_X(0) \quad \text{indip. da } n_i$$

$$R_X(0) = \sigma_X^2$$

$$C_{i,i} = 1$$



$$n_{i+1} - n_i = P \quad \text{è indep. da } i$$

$$\underline{q}^T \underline{X} = [q_1 \quad q_2 \quad \dots \quad q_m] \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_m \end{bmatrix} = q_1 X_1 + q_2 X_2 + \dots + q_m X_m$$

$$E[\sum_{i=1}^n c_i X_i] = E[c_1 X_1 + \dots + c_n X_n] = 0$$